

CM6453 SW Processes Term Project

Distributed Agents for the Software Processes Course

In this term project, you will develop a web-based software system using a **distributed multi-agent architecture**.

Each computer will run an LLM-based agent. A **master agent** will be responsible for decomposing programming tasks into smaller subtasks and distributing them among the available agents. The agents may use different LLM models with different capabilities. Therefore, the master agent should evaluate the capabilities of each available model and assign tasks according to the strengths and limitations of the individual agents.

In the software development process you will follow the **Scrum methodology**. By using this methodology you will gain practical experience with agile software development processes, including sprint planning, task assignment, incremental development, testing, and sprint reviews.

You will use Trello for planning and task management .

You should create a github repository and keep your code and documentation in this repository.

The project will consist of **five iterations (sprints)**. Each iteration will last **two weeks**, resulting in a total development period of **10 weeks**. At the end of each iteration, the system should provide a functional increment of the overall product.

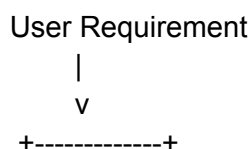
The distributed-agent architecture should be an integral part of the project. You should demonstrate how the master agent decomposes tasks, evaluates agent capabilities, distributes tasks, coordinates their execution, and integrates the results.

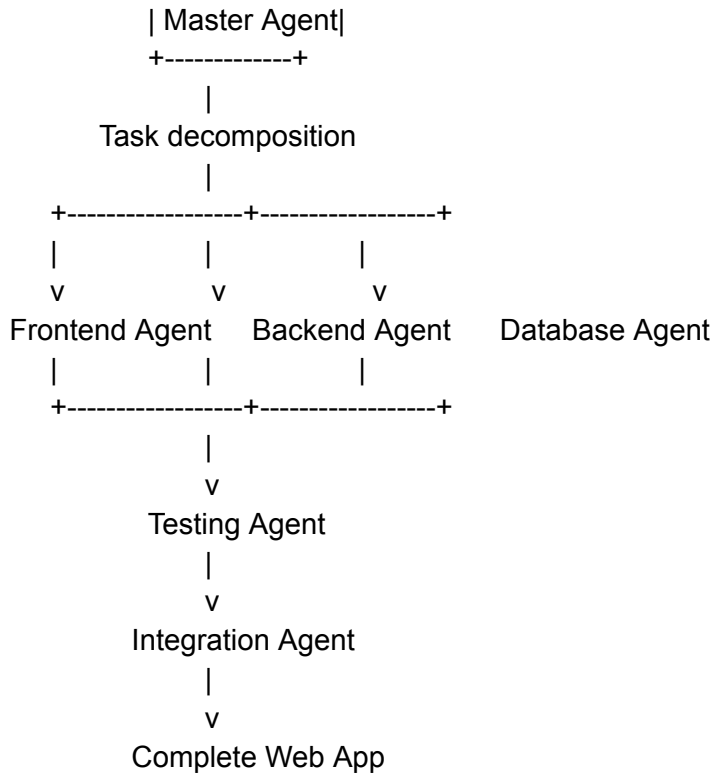
Distributed AI Software Development Platform

The idea is that a user enters a software requirement such as:

"Develop an online library management system." (pls note that the the actual requirement will be long document not a single sentence)

The master agent analyzes the requirement and decomposes it into tasks:





The interesting part for your course is that the **master agent does not simply distribute tasks randomly**.

For example:

Agent A

Model: GPT-X

Coding capability: 90

Reasoning capability: 85

Context capacity: 95

Agent B

Model: Model-Y

Coding capability: 75

Reasoning capability: 95

Context capacity: 70

Agent C

Model: Model-Z

Coding capability: 95

Reasoning capability: 70

Context capacity: 80

The master agent could therefore assign:

Requirements analysis → Agent B
Frontend development → Agent C
Backend development → Agent A
Testing → Agent A/B
Documentation → Agent B

Possible 5-Sprint Plan

Sprint 1 — Foundation and Requirements

- Define system requirements
- Design overall architecture
- Implement web interface
- Implement master-agent prototype
- Register available agents/models
- Implement basic task decomposition

Sprint 2 — Agent Management and Task Assignment

- Agent capability profiles
- Model capability evaluation
- Task classification
- Agent selection algorithm
- Task distribution
- Agent status monitoring

Sprint 3 — Distributed Development

- Agents execute programming tasks
- Communication between master and agents
- Shared project repository
- Result collection
- Dependency management between tasks

Sprint 4 — Integration and Quality Assurance

- Code integration
- Automated testing
- Code review agent
- Error detection and recovery
- Conflict resolution
- Reassignment of failed tasks

Sprint 5 — Complete System and Evaluation

- End-to-end workflow
- Dashboard
- Performance measurements
- Agent/model comparison
- Documentation
- Final testing and demonstration

This project also gives you a very good opportunity to assess **Scrum practices**: each sprint can require a product increment, sprint backlog, task assignments, review/demo, retrospective, and progress report.